PSEL4 is a drop-in auto level for pytket-quantinuum

- PSEL4 is a cost-aware auto level. It compiles each circuit four ways and keeps the cheapest checked result.
- It is never costlier than the default compile (0 of 217 circuits).
- It needs 15.7 % less HQC than the default level 2 on our 217-circuit benchmark.
- **[O]** Workload assumption: a team spends the Standard Plan's 10,000 HQC a month on this mix. PSEL4 then frees 1,573.3 HQC, worth USD 19,666 a month.

Team Learned ZX Rewriting

C.-F. Kam, A. Mahasinghe, D. De Silva, N. Karunarathne, F. Cadet, M. Bessafi, K. De Silva

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An H2 user pays HQC for every compile choice

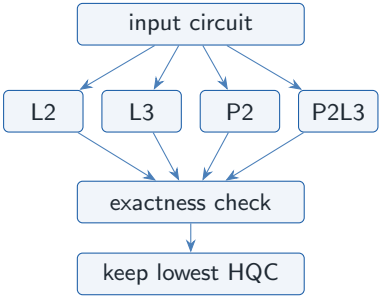
$$\mathrm{HQC} = 5 + C\left(N_{1q} + 10\,N_{2q} + 5\,N_{\mathrm{SPAM}}\right)/5000$$

C is the shot count. N_{1q} counts PhasedX gates, N_{2q} counts ZZPhase and ZZMax gates and N_{SPAM} counts initialisations, measurements and resets. Rz gates are free.

- **User.** A team that runs circuits on H2 under a fixed HQC budget.
- pytket-quantinuum compiles at optimisation level 2 by default.
- Always using level 3 saves 13.9 % HQC, yet costs more than level 2 on 45 of 217 circuits.
- Blind ZX rewriting (PyZX teleport_reduce) before level 2 costs 6.27 % more HQC on 180 Trotter circuits.
- **[O]** The user cannot tell in advance which compile is cheapest for a given circuit.

PSEL4 compiles four ways, checks each and keeps the lowest HQC

- **Candidates.** L2 (default), L3, and PyZX teleport_reduce followed by L2 (P2) or L3 (P2L3).
- **Check.** Each candidate is checked against its input as a full unitary up to 8 qubits and on 64 random computational basis inputs (including all-zero) from 9 to 20 qubits. Failed candidates are dropped.
- Above 20 qubits family C relies on PyZX miter proofs.
- **Guarantee.** L2 and L3 are always candidates, so PSEL4 never costs more than either where both pass the check.



Drop-in use. One command per circuit file, printing every candidate's HQC and the pick.

```
python3 pipelines/pass_select.py --psel4 <circuit.qasm>
```

Finding 1. Level 3 adoption made safe

Policy	Total HQC	Saved vs L2	Circuits costlier than L2	HQC lost there
Always L2 (default)	58,837.2			
Always L3	50,632.2	13.9 %	45	372.0
PSEL4	**49,580.6**	**15.7 %**	**0**	**0.0**

217 circuits (A 180, B1 15, C 20, Grover use case 2), HQC at 1000 shots. Source `value/policy_compare.md`, `value/usd_psel4.md`.

- Split of the 15.73 % saving: always level 3 gives 13.95 points, per-circuit selection plus ZX give 1.79 points.
- Against the per-circuit cheaper of level 2 and level 3, PSEL4 saves 679.6 HQC (1.35 %).
- Always level 3 costs more than level 2 on 45 circuits, up to 12.95 % on one (gf2_pow_4_mult). PSEL4 on none.
- The largest saving is on family C (arithmetic and oracles), 25.8 % below level 2.
- Four compilations take 5.08 times the wall time of level 2 (215 timed circuits), or 3.03 HQC saved per extra second.

Finding 2. The HQC-cheapest compile is not the most accurate

Circuit	Pick	p level 2	p pick	z	R
cdkm_adder_10	L3	0.8692	0.8267	-2.90	0.9203
hwb6	P2L3	0.9200	0.8883	-2.64	0.8909
modadd_2n_10	P2L3	0.8417	0.8383	-0.22	0.9526
vbe_adder_3	L3	0.9531	0.9569	$+0.51$	0.6713
vbe_adder_7	L3	0.8830	0.9010	$+1.30$	0.6114
Pooled per answer, 8 pairs					**0.8192**

Pre-registered H2-2E runs of the PSEL4 pick against level 2 on non-zero inputs, 3 to 4 each at 400 shots per program. p: share of correct shots. z: pooled two-proportion statistic. R: HQC per correct output, pick over level 2. The pooled row adds three earlier P2 pairs. Source `value/hqc_per_correct.md`.

- The pick was significantly less accurate on 2 of 5 (hwb6 Holm $p = 0.0168$), not different on 2, not significantly better on 1. No pick was significantly more accurate.
- On cdkm_adder_10 the less accurate pick has fewer two-qubit gates (118 against 141) and lower two-qubit depth (65 against 122, post hoc), so neither HQC nor gate counts order fidelity.
- HQC per correct answer is lower for the pick in 5 of 5 pairs, 8 of 8 with the earlier pairs. Pooled 18.1 % lower (bootstrap 95 % interval 0.8132 to 0.8254), HQC ratio 0.8170 over accuracy ratio 0.9973.
- **[O]** PSEL4 serves cost-bound users. Accuracy-bound users need the fidelity-aware finals selector.

Value of the saving in USD

- Azure plan averages are USD 10.29 (Premium) to 12.50 (Standard) per HQC.
- Against always level 2, PSEL4 saves 9,256.6 HQC, worth USD 95,289 to 115,707.
- Against always level 3, PSEL4 saves 1,051.6 HQC, worth USD 10,825 to 13,145.
- Always level 3 loses USD 3,829 to 4,650 on its 45 costlier circuits. PSEL4 loses nothing there.
- **[O]** Workload assumption: a team spends the Standard Plan's 10,000 HQC a month on this circuit mix. PSEL4 then frees 1,573.3 HQC a month, worth USD 19,666 at the Standard average.

Ledger: 217 circuits, 1000 shots each, HQC formula estimate. Plan averages are not marginal prices and ignore bundled eHQCs and Azure infrastructure costs. Source `value/usd_psel4.md`.

Finding 3. Level 3 cuts two-qubit gates and ZX cuts phases

	Non-Clifford gates				Two-qubit gates			
Set	L2	P2	L3	P2L3	L2	P2	L3	P2L3
B1, 14 circuits	877	656	746	701	862	799	723	700
C, 15 circuits	3323	2930	2540	2535	3208	2944	2718	2708

Totals after each pipeline. 6 circuits did not fit the compile budget. Source `value/mechanism_zx_l3.md`.

- **Mechanism.** HQC charges no Rz gate and bills every PhasedX at the same rate, so the T-count reduction ZX is best at has no HQC value on H2, and ZX can pay only through two-qubit gates, which level 3 already resynthesises.
- GreedyPauliSimp in level 3 lowers the two-qubit count on 180 of 180 A, 13 of 14 B1 and 9 of 15 C circuits.
- teleport_reduce leaves fewer non-Clifford gates on B1 (656 against 746 for level 3). A two-qubit gate costs 2 HQC at 1000 shots and a PhasedX 0.2.
- ZX on top of the cheaper pytket level saves 3.21 % on all 20 family C circuits (0.26 % on the 17 proven), 0.49 % on B1 and 0 on Trotter circuits. Its gain anticorrelates with the level 3 gain, $\rho = -0.64$.
- On B1 the ZX stage removes 40.6 % of T gates. The Gosset et al. (2014) bound leaves at most 0.3 % for further exact T optimisation (24 circuits of at most 5 qubits).

17 of 20 PSEL4 outputs on family C are proven equal to their input

Family C output	Proven	Not proven	Saved vs L2 (proven)
Level 3	18 of 20	hrs_mult_5, rgqft_mult_6	13.33 %
PSEL4	17 of 20	hrs_mult_5, rgqft_mult_6, draper_adder_28	10.22 %

Family C has 20 to 56 qubits. A PyZX miter reduction proves equality of the full unitary up to global phase. Sources `results/family_C_equivalence_L3.md`, `value/psel4_zx_only.md`, `value/psel4_basis_check.md`, `value/psel4_conservative.md`, `results/family_C_equivalence_qcec.md`.

- On families A, B1 and B2 every compiled circuit matches the uncompiled reference within 1e-8.
- 64 random basis inputs, 9 to 20 qubits: all 83 PSEL4 picks pass, with 187 of 187 required and 142 of 142 other candidates. Negative controls fail as they should.
- MQT QCEC 3.10.2 proved none of the three unproven PSEL4 outputs. Its verdicts were inconclusive, with no counterexample on any pair.
- Conservative variant (cheapest verified candidate, else level 2 flagged "default, unverified"): 12.25 % below level 2. It is costlier than level 2 on 0 circuits and than level 3 on 2 whose level 3 output is unverified.

Limitations

- HQC is the formula estimate. All fidelity results come from the H2-2E emulator. Nothing has run on H2 hardware.
- The PSEL4 pick was significantly less accurate on 2 of 5 pre-registered pairs (cdkm_adder_10 $z = -2.90$, hwb6 $z = -2.64$). Accuracy-bound users need a fidelity-aware choice.
- Per-answer intervals pool inputs and ignore between-run variation. 3 of 8 per-answer pairs use P2, the earlier selector's pick.
- The ZX-only gain is 0.26 % on the 17 proven family C circuits.
- 2 family C level 3 outputs and 3 PSEL4 outputs are unproven. On draper_adder_28 the PSEL4 pick starts from angles PyZX rounded by up to 1.50e-06 rad.
- The earlier vbe_adder_3 fidelity gain used the all-zero input. Its Holm-adjusted p is 0.052 over three circuits.
- No user interview or survey is part of this submission. The selector follows pytket's PassSelector pattern.

Finals plan is a fidelity-aware learned selector

- **[O]** Goal: a learned selector that predicts fidelity as well as HQC for each candidate. It keeps the per-answer saving and protects accuracy where users need it.
- Finding 2 shows why. On 2 of 5 pre-registered pairs the HQC-cheapest compile returns the correct output less often.
- Submission 1 provides the training data: the checked ledger (217 circuits, 4 candidates each) and the emulator labels (5 pre-registered PSEL4-pick pairs, 48 H2-2E jobs).
- A first classifier on input-circuit features does not beat a constant choice with a whole family held out (balanced accuracy 0.433 against 0.500).
- **[O]** The model therefore needs compiled-candidate structural features and emulator labels.

Every result can be rerun from the code package

- `code.zip`. The source of the repository. `requirements.txt` pins pytket 2.18.4, pytket-quantinuum 0.59.3 and pyzx 0.10.7.
- `run_tests.sh`. Runs 38 test files, including the 64-basis-input check and its negative-control test, and writes one return code per file.
- **Notebook.** Demonstrates PSEL4. It compiles tof_3 with PSEL4 and submits the pick to the H2-2SC syntax checker.
- `EVIDENCE.md`. Lists every emulator job with its job ID, shots, HQC estimate and payload. The 48 H2-2E jobs used an HQC estimate of 9971.8 of the 10,000 budget.
- **Generated tables.** Value tables come from the scripts under `value/`. Emulator tables are regenerated offline from the stored payloads.
- **Independent checks.** Verification reports by agents that did not produce the results are in `results/`.